



Internship Report Form

Student-Intern's Name	:	Skyler Jason P. Banzon		
Name of HTE	:	Cubeworks Technology Consulting and Solutions, Inc.		
HTE Supervisor's Name	:	Elmer N. Canque (CIO)		
Report Period	From	: March 16, 2026	To	: April 15, 2026
Number of Hours Worked This Quarter	:	160		
Cumulative No. of Hours to Date	:	240		

Quarter Summary

No.	Tasks Performed	% Time Spent
1	Continuation and completion of Node-RED practice simulations: finalized remaining Factory I/O machine sequences including a parking lot automation demo, a heater control system, and a traffic light controller with 3D JavaScript visualization and database logging	8%
2	Informed by seniors and supervisors to transition from Node-RED and Factory I/O training to actual ongoing internal projects; received overviews of each project's pipeline and selected a project track of their choosing.	7%
3	Assigned to the wire harness measurement project; studied pipeline requirements, set up the development environment, and explored ArUco markers as a spatial reference for real-world scale measurement.	10%
4	Built an initial functional prototype using an ArUco-based pipeline and IP Webcam stream; implemented core image processing and skeletonization to measure wire strand lengths in real time.	17%
5	Improved measurement stability by introducing temporal smoothing across frames; added per-strand confidence indicators and refactored the codebase for more reliable strand tracking between frames.	13%
6	Improved spatial accuracy by overhauling the coordinate system to use A4 paper dimensions as a reference; refined marker masking logic and added a visible measurement zone boundary overlay.	13%
7	Extended the system to support Y-shaped wire harnesses by detecting skeleton junctions and separating the wire into a trunk and branches; added a dedicated Measurements window and a keyboard-toggled fallback mode.	17%
8	Redesigned the branch detection system to support any number of branches and junctions; improved robustness of trunk identification, smoothing, and branch visualization through a full codebase refactor.	15%
TOTAL		100%



Additional Activities (other growth opportunities e.g., conferences, training, seminars etc.)

- Used prior knowledge from my thesis in using computer vision concepts such as homography and learned new techniques like adaptive thresholding, skeletonization, and graph-based skeleton traversal. Reviewed OpenCV ArUco and scikit-image documentation to better understand the algorithms used in the project.

List any holidays, personal days, and/or days missed this quarter:

- Observed national holidays during the reporting period: Maundy Thursday, Good Friday, Black Saturday, and Day of Valor (April 2, 3, 4, and 9). Full attendance on all other working days.

Quarter Analysis, Reflection, and Plan of Action

What experiences were particularly rewarding during this reporting period?

Being assigned to an actual ongoing project with a clear real-world purpose was a meaningful shift from training. Watching the wire measurement system evolve iteratively from v1 to v5 was a satisfying demonstration of incremental software development.

What experiences were particularly difficult during this reporting period?

Achieving consistent wire segmentation under varying lighting conditions required careful and sensitive parameter tuning. The branch detection work in v4 and v5 introduced significant complexity around junction handling and skeleton graph analysis.

(If Applicable) Other comments, items, or issues that you want your supervisor or adviser to be made aware of? What is your proposed plan of action to resolve this issue?

No major issues to report; the pipeline is functional and progressing steadily. Focus going forward is on improving segmentation robustness and exploring potential integration with the team's broader ongoing systems.



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School of Engineering


Skyler Jason P. Banzon
Student-Intern


Elmer N. Canque
HTE Supervisor

Date Submitted : April 15, 2026

Date Reviewed : April 15, 2026

Supervisor's Comments:

None